Sri Sivasubramaniya Nadar College of Engineering, Chennai

(An autonomous Institution affiliated to Anna University)

Degree & Branch	BE Computer Science & Engineering	**Semester**	VI
Subject Code & Name	Machine Learning Laboratory		
Academic Year	2025–2026	**Batch**	2023–2027
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Experiment 9 Report

Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering

Aim and Objective

The objective of this experiment is to implement and compare three clustering algorithms on the Human Activity Recognition (HAR) dataset:

- **Model A:** K-Means Clustering
- **Model B:** DBSCAN (Density-Based Spatial Clustering of Applications with Noise)
- **Model C:** Hierarchical Agglomerative Clustering (Ward linkage)

The models are evaluated through internal and external clustering metrics, and interpreted visually through PCA/t-SNE projections.

Dataset Description

Dataset: Human Activity Recognition Using Smartphones Dataset (UCI)

- 30 volunteers aged 19–48 performed six activities.
- Activities include six classes spanning walking and postural actions.
- Sensor signals from tri-axial accelerometer and gyroscope sampled at 50Hz.
- Preprocessed features were used from the provided train and test files.

Data used in notebook:

- Combined feature matrix shape: **(10299, 561)**
- Missing values: **0**

Preprocessing Steps

1. Loaded feature names from `features.txt` and activity labels from `activity_labels.txt`.
2. Loaded training and testing sets, then concatenated them into a single dataset.
3. Mapped numeric class labels to activity names for interpretability.
4. Standardized all features using `StandardScaler` (zero mean and unit variance).
5. Applied dimensionality reduction:
   - PCA to 2D for visualization
   - PCA to 50 dimensions for clustering efficiency
   - t-SNE (2D) on PCA-50 features for visual analysis

K-Means with Elbow Method Results

K-Means was run for $k = 2$ to $k = 8$ on PCA-50 features. WCSS (inertia) and silhouette scores are tabulated below.

Table 1: K-Means Elbow Method Results

Number of Clusters (k)	**WCSS (Inertia)**	**Silhouette Score**
2	2527535.19	0.4501
3	2176124.28	0.3678
4	2036867.90	0.1903
5	1911005.10	0.1662
6	1833594.70	0.1524
7	1776514.72	0.1227
8	1722452.60	0.1162

Best k selection:

- Elbow and silhouette trends indicate strongest global separation at $k = 2$.
- This corresponds to two broad groups: dynamic vs static activities.
- For activity-wise comparison with true labels (6 classes), the final K-Means model was run with $k = 6$.

Clustering Implementation

Model A: K-Means

- Configuration: `n_clusters=6`, `random_state=42`, `n_init=10`
- Input space: PCA-50 features

Model B: DBSCAN

- Configuration: $\epsilon = 14.0$, `min_samples=10`
- Input space: PCA-50 features
- Clusters found: **2**
- Noise points detected: **660 out of 10299**

Model C: Hierarchical Agglomerative Clustering

- Linkage method: **Ward**
- Dendrogram built on a random sample of 1000 points for computational feasibility
- Final clustering: `n_clusters=6`

Evaluation Metrics and Results

The following metrics were computed:

- **Internal:** Silhouette Score, Davies–Bouldin Index, Calinski–Harabasz Index
- **External:** Adjusted Rand Index (ARI), Normalized Mutual Information (NMI)

Table 2: Comparison of Clustering Metrics

Algorithm	**Silhouette**	**Davies–Bouldin**	**Calinski–Harabasz**	**ARI**	**NMI**
K-Means ($k = 6$)	0.152444	1.968146	3591.285346	0.420093	0.559792
DBSCAN	0.324938	1.435599	805.265184	0.013575	0.060240
Hierarchical (Ward, $k = 6$)	0.129356	2.064510	3292.688217	0.421752	0.579220

Graphs and Visualizations

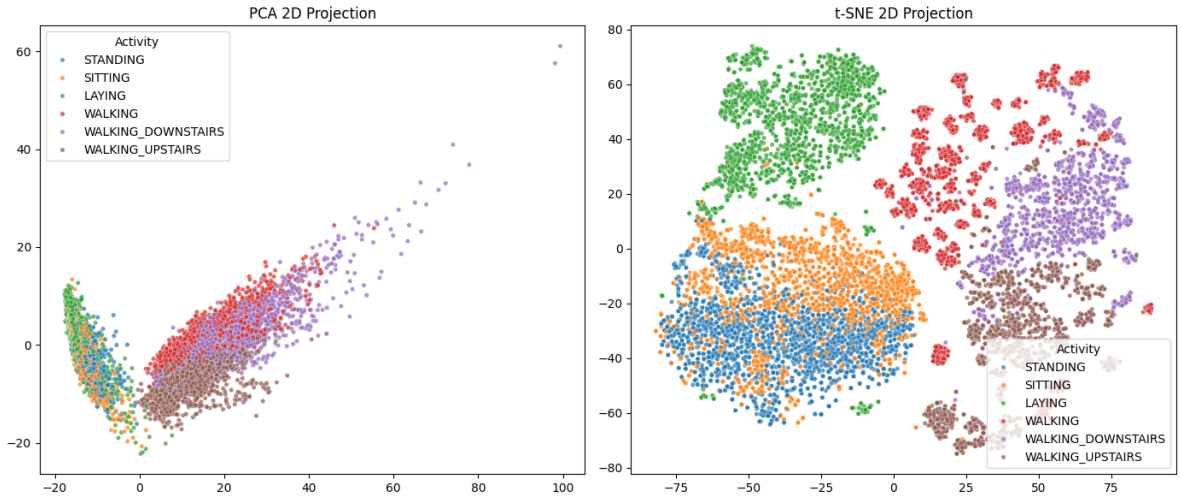

Figure 1: PCA and t-SNE projections colored by true activity labels.

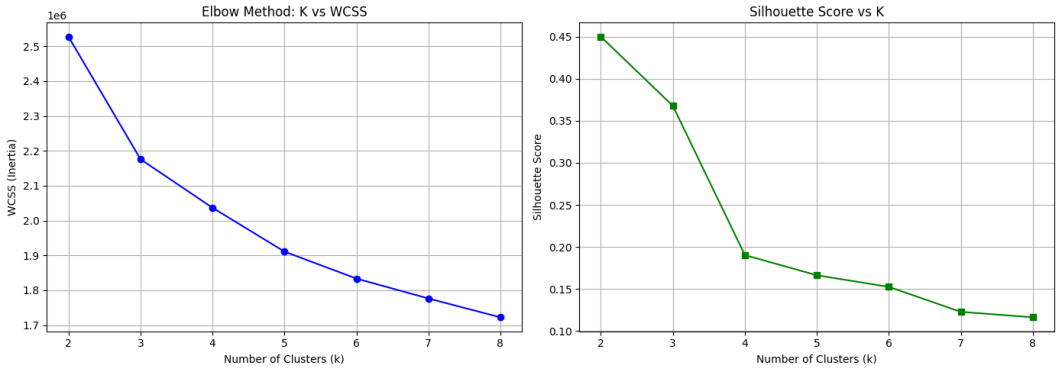

Figure 2: Elbow curve (WCSS vs k) and silhouette score vs k.

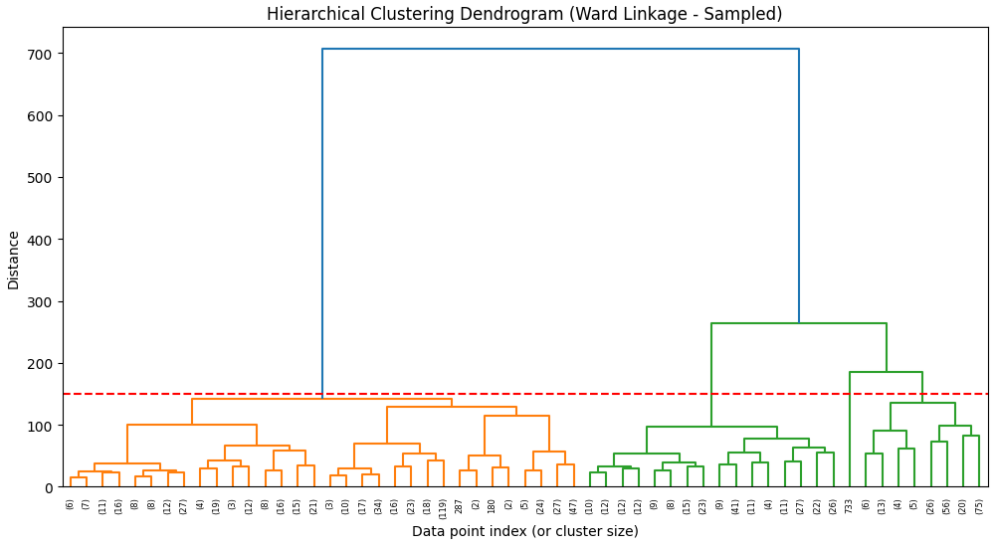

Figure 3: Hierarchical clustering dendrogram (Ward linkage, sampled points).

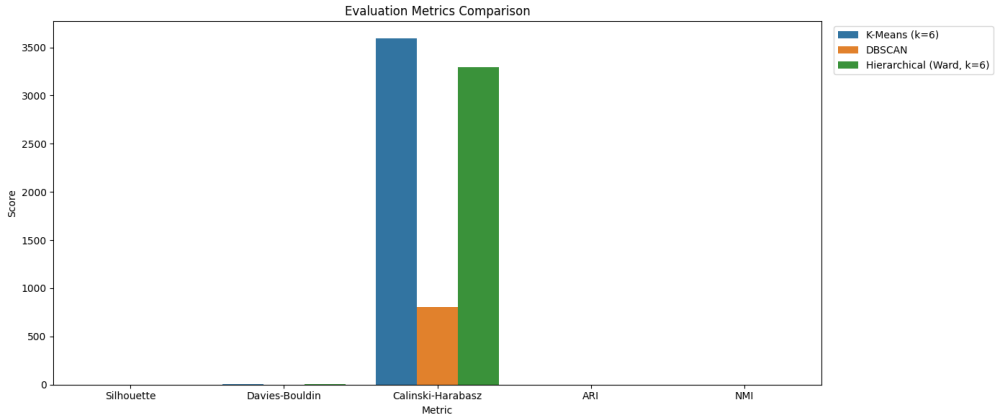

Figure 4: Bar plot comparison of evaluation metrics across clustering algorithms.

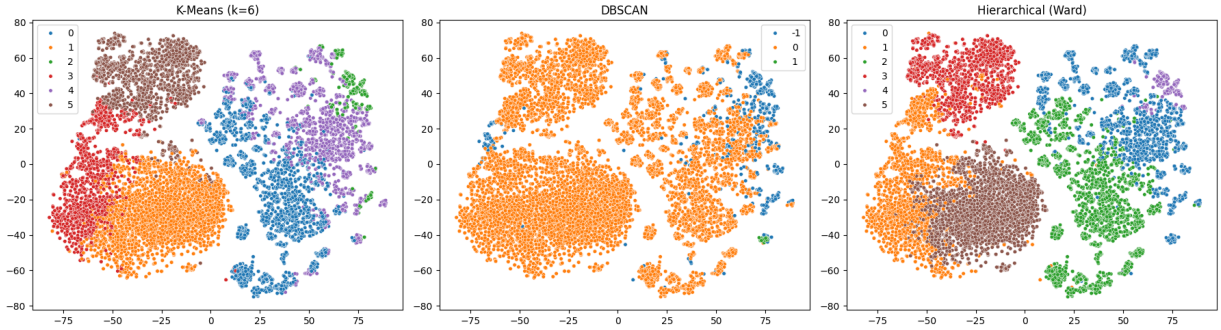

Figure 5: t-SNE scatter plots of cluster assignments from K-Means, DBSCAN, and Hierarchical models.

Observations and Comparative Analysis

1. **Most meaningful clusters:** K-Means ($k = 6$) and Hierarchical Ward ($k = 6$) gave meaningful activity-level groupings. Their ARI and NMI values were substantially higher than DBSCAN and were close to each other.

2. **Sensitivity of K-Means to k:** K-Means was sensitive to k. The silhouette score was highest at $k = 2$, while activity-wise clustering required $k = 6$, which lowered silhouette due to overlap among similar classes (especially static postures).

3. **DBSCAN behavior:** DBSCAN detected noise points (660), but failed to align clusters with true activity labels (very low ARI and NMI). This indicates limited suitability here under the chosen density settings.

4. **Effect of linkage choice:** Ward linkage provided compact and balanced clusters. In general, single linkage may suffer from chaining, while complete linkage may be more sensitive to outliers.

5. **Metric matching visual intuition:** Silhouette score best reflected broad cluster separability in t-SNE space (favoring 2 macro-clusters), while ARI/NMI better reflected label-level alignment for the 6 activities.

Conclusion

This experiment demonstrates that clustering quality depends strongly on objective: global separation vs fine-grained activity separation. K-Means and Hierarchical Ward were most effective for six-class comparison, while DBSCAN was comparatively weak for this transformed feature space. Combining quantitative metrics with PCA/t-SNE visualization gave a clear and balanced interpretation of cluster behavior.